Skills Summary

Computing and Interpreting Centre and Spread

Use this reference while completing your worksheet or when studying.

Measures of Centre: Mean, Median, Mode

Mean = total ÷ how many. **Median** = the middle value once the list is in order (the mean of the middle two when there is an even count). **Mode** = the value that appears most often.

Order the list first, every time. An unordered list makes the median a guess.

Example: Find the mean and the median of 3, 9, 5, 10, 8.
Step 1. The mean needs the total and the count; the median needs the list in order, so do the ordering before anything else.
Step 2. Total = 35, so the mean is $35 \div 5 = \mathbf{7}$; in order the list is $3, 5, 8, 9, 10$, so the median is the third value, **8**.

Try it: Find the median of 4, 11, 6, 15, 9, 7.

check: **8**

Measures of Spread: Range and Interquartile Range

Range = largest − smallest. It uses only the two ends, so a single unusual value changes it completely.

Interquartile range = upper quartile − lower quartile. Order the list, split it into a lower half and an upper half (leave the middle value out if there is one), and take the median of each half. It measures the middle half, so the ends cannot move it.

Example: Find the quartiles and the interquartile range of 5, 7, 9, 12, 15, 18, 20, 24.
Step 1. Eight values in order split evenly into a lower four and an upper four, so take the median of each half.
Step 2. Lower half $5, 7, 9, 12$ gives **8**; upper half $15, 18, 20, 24$ gives **19**; the interquartile range is $19 - 8 = \mathbf{11}$.

Try it: Find the interquartile range of 1, 5, 9, 13, 17, 21, 25, 29.

check: **16**

Choosing the Pair that Tells the Truth

Rule: report a centre *and* a spread, and match them.

- Values fairly even, no extremes: **mean with range** is fine.
- One value far from the rest: **median with interquartile range**, because neither is dragged by that one value.

Example: Five prices, in dollars: 4, 5, 5, 6, 30. Which centre describes them?
Step 1. One value sits far above the others, so compute both the mean and the median and see whether they agree.
Step 2. The total is 50, so the mean is $50 \div 5 = \mathbf{10}$, while the median is **5** — and four of the five prices are under 7, so the median is the honest one.

Try it: Find the median of 2, 3, 3, 4, 38, and say whether it or the mean describes these values better.

check: median **3**, and it is the better description

(!) Watch out: you cannot average two means when the groups differ in size. For 4 students at 80 and 6 at 90, recover the totals first: $320 + 540$ over 10 students gives 86, not 85.